CLINICAL TRIAL FAILURES · MISSED ENDPOINTS · RESPIRATORY PHASE 2,3 · STARTS 2015–2024

# Alarmins (TSLP / IL-33): 5 of 8 completed trials missed their primary endpoint

**5 completed trials across 3 sponsors and 4 molecules ran to the end and missed their primary endpoint on the sponsor's own posted result; 3 met it.** Only 0 of 24 closed trials were stopped early for efficacy or safety, which is why this class is read through its completed trials rather than a stop rate. Most completed trials post nothing readable, so the count is a floor.

**5 of 8**
completed trials with a readable primary result missed it · 3 met it

**3 / 4**
sponsors / molecules behind the misses

**0.0%**
stopped early: 0 of 24 closed trials (95% CI 0.0%–13.8%)

**77.4%**
of 31 trials have closed · 7 still open or unresolved

**5 of 8** **Completed and missed the primary endpoint.** Of 23 completed trials in this segment, 8 posted a primary result on ClinicalTrials.gov that can be read — the sponsor's own comparison held to the sponsor's own threshold, or its statement that the endpoint was missed: 5 missed and 3 met. Counted separately and never in the rate above; a floor, not a rate. 1 of the 20 completed trials that finished more than a year ago have posted no results at all — a gap in what can be known, not a sign of failure.

## The completed trials that missed

| Trial | Ph | Sponsor | Experimental drugs | Started | What the posted result says |
|---|---|---|---|---|---|
| NCT06280391 | 2 | Sanofi | ITEPEKIMAB | 2024-02 | Annualized Rate of Adjudicated Moderate or Severe Pulmonary Exacerbations (PEs) Over the Treatment Period — p 0.3419, no threshold stated; the sponsor posted a 90% interval, i.e. a design tested at two-sided 0.1: not significant. Rate ratio 0.851 (90% CI 0.643 to 1.126). Method: Negative binomial regression model. |
| NCT05037929 | 2 | Roche | ASTEGOLIMAB | 2021-10 | Annualized Rate of Moderate and Severe COPD Exacerbations Over the 52-Week Treatment Period — p 0.3829, no threshold stated; 0.2 or above, so not significant under any conventional threshold. Adjusted Rate Ratio 0.93 (95% CI 0.789 to 1.095). Method: Negative Binomial Regression Model. |
| NCT04751487 | 3 | Sanofi | ITEPEKIMAB | 2021-02 | Former Smoker Cohort: Annualized Rate of Adjudicated Moderate or Severe Acute Exacerbation of Chronic Obstructive Pulmonary Disease (AECOPD) Events Over the 52-Week Placebo-Controlled Treatment Period — p 0.2176 against the sponsor's stated threshold of 0.049: not significant. Rate ratio 0.876 (95% CI 0.71 to 1.081). Method: Negative binomial regression model. |
| NCT04386616 | 2 | Roche | ASTEGOLIMAB, EFMARODOCOKIN ALFA | 2020-06 | Time to Recovery, Defined as the Time to a Clinical Status Score of 1 or 2 on the 7-Category Ordinal Scale (Whichever Occurs First) by Day 28 — p 0.9266, no threshold stated; 0.2 or above, so not significant under any conventional threshold. Hazard Ratio (HR) 1.01 (95% CI 0.75 to 1.36). Method: Unstratified Log Rank. |
| NCT03406078 | 3 | AstraZeneca | TEZEPELUMAB | 2018-03 | Categorized Percent Reduction From Baseline in the Daily OCS Dose While Not Losing Asthma Control — p 0.434, no threshold stated; 0.1 or above, so not significant under any conventional threshold. Cumulative Odds Ratio 1.28 (95% CI 0.69 to 2.35). Method: Proportional odds model. |

## How this compares

Alarmins (TSLP / IL-33)
0.0%
0/24 closed

all respiratory trials with a resolved drug target
4.2%
35/840 closed

All respiratory Phase 2,3
4.2%
82/1939 closed

**What is in this class:** 9 distinct experimental drugs across 31 trials (antibody 31, protein 1, small molecule 1). Most tested: TEZEPELUMAB (17), ITEPEKIMAB (7), ASTEGOLIMAB (6), ETOKIMAB (1), EFMARODOCOKIN ALFA (1). A class groups drugs by what they act on; check that the grouping is one you would make before reading the rate as a property of the mechanism.

**How much does this rest on one decision?** The 0 stops came from 0 sponsor-asset programmes across 0 sponsors. Screened alongside 13 other segments, the rate is within what screening that many classes could produce (q=1); the molecules and trials that failed here are the record either way. Read it as a screen worth checking against the underlying trials.

Bars show the share of closed trials stopped for biological reasons. Confidence intervals overlap where sample sizes are small — read the counts, not just the bars.

## By start cohort

| Trials started | Stops | Closed | Rate | 95% CI |
|---|---|---|---|---|
| 2018–2020 | 0 | 11 | 0.0% | 0.0%–25.9% |
| 2021–2024 | 0 | 9 | 0.0% | 0.0%–29.9% |

## Where the stops sit

.
Sponsors with most stops: .

**Method**
Denominator: ClinicalTrials.gov interventional Phase 2,3 respiratory trials started 2015–2024 that have closed (completed or terminated). Numerator: terminated trials whose registry stop reason is classified as biological (efficacy, safety or benefit–risk) by Classification V2 — held-out precision 95.5%, recall 95.3% (n=600). Drugs are linked to ChEMBL and the NCI Thesaurus; 41% of industry respiratory trials in this window carry a resolved drug target, and a trial without one cannot enter a mechanism class. Intervals are Wilson 95%.

**Limits**
Not a failure rate: completed trials that missed their endpoints are not in the rate. They are counted separately where the sponsor posted a primary result that can be read — most did not, so that count is a floor — and programmes discontinued after a completed trial do not appear. Stop reasons are sponsor-reported. This is a closed-trial proportion, not a time-to-event analysis: only 77.4% of trials in this segment have closed, and a trial that stops early enters the denominator sooner than one that runs to completion, which can inflate the rate in immature segments. Recent cohorts have fewer closed trials, so their rates are less stable. Research signals, not clinical or investment advice.

**The whole cohort, trial by trial.** The diligence report names every trial behind these figures — the ones still running and the ones that completed and missed their primary endpoint included — and places your molecule against each one that failed. €99, delivered immediately: **clinicaltrialfailures.com/asset-check**

Sources: ClinicalTrials.gov (NLM); ChEMBL (EMBL-EBI, CC BY-SA 3.0); NCI Thesaurus (NCI); RxNorm/RxClass (NLM); SEC EDGAR. Classification, linkage and rates by Clinical Trial Failures. Rebuilt weekly; this brief covers trials started 2015–2024.